



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE	TECH WORKFORCE SALARY & FINANCIAL ANALYTICS DASHBOARD
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PROJECT CATEGORY	• Education Analytics • Social Media Analytics • Mental Health Analytics • Student Lifestyle Analytics
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INSTITUTE NAME	Infinity Code Nexus
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TRAINER / GUIDE	Mr. Anand Kumar Anamaina
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CREATED BY	Vishnu Bandari
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TOOLS USED	Power BI Desktop Power Query, DAX & Power BI Service
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SUBMISSION DATE	[09/06/2026]
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Prepared as part of the Power BI Masterclass Program to demonstrate skills in Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX, Storytelling, and Business Intelligence Reporting.

PROJECT OVERVIEW

PROJECT OBJECTIVE

Develop a Power BI dashboard to analyze the impact of social media usage on student mental health, stress levels, sleep quality, study habits, and lifestyle factors.

Participants are expected to:

- ✓ Analyze social media usage patterns
- ✓ Measure mental health scores
- ✓ Evaluate stress levels
- ✓ Analyze sleep quality
- ✓ Understand study habits
- ✓ Generate meaningful insights for student well-being

PROJECT GOAL

Transform student behavioural and social media data into actionable insights that help educators and researchers understand factors affecting mental health and academic performance.

BUSINESS PROBLEM

Students spend significant time on social media platforms. Excessive usage may influence stress levels, sleep quality, study hours, and overall mental health.

The objective is to identify:

- Which social media platforms are most used?
- How social media usage affects mental health?
- Does sleep quality influence stress levels?
- How study hours compare with social media usage?
- Which countries show better mental health scores?

BUSINESS QUESTIONS

Participants must answer:

1. What is the average mental health score?
2. Which social media platform is used most frequently?
3. How many students belong to each stress category?
4. Does sleep quality affect mental health?
5. Which academic level has the highest mental health score?
6. Which countries have better mental health outcomes?
7. How does physical activity impact mental health?

DATASET INFORMATION

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DATASET NAME : Student Social Media Mental Health Dataset

DATASET DETAILS : 5,000 Records

Columns : 18

EXPECTED ANALYSIS AREAS

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Social Media Analytics
Mental Health Analytics
Stress Analytics
Sleep Analytics
Academic Analytics
Lifestyle Analytics
Country-wise Analysis

POWER QUERY TRANSFORMATIONS

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DATA CLEANING ACTIVITIES

- ✓ Removed Duplicate Records
- ✓ Handled Missing Values
- ✓ Corrected Data Types
- ✓ Standardized Categories
- ✓ Removed Unnecessary Columns

DATA PREPARATION ACTIVITIES

- ✓ Renamed Columns
- ✓ Created Custom Columns
- ✓ Replaced Values
- ✓ Data Formatting
- ✓ Data Validation

CUSTOM COLUMNS CREATED

Stress Category

Sleep Category

Usage Category

Study Hour Group

Health Category

SCREENSHOT: Power Query

fx

= Table.AddColumn("#Added Custom3", " Stress_Order", each if [Stress_Level] = "Low" then 1

	1.2.3 Age	1.2.4 Gender	1.2.5 Country	1.2.6 Academic_Level	1.2.7 Most_Used_Platform	1.2.8 Purpose_Of_Use	1.2.9 Avg_Daily_I
	Valid Error Empty 100%0%0%	Valid Error Empty 100%0%0%	Valid Error Empty 100%0%0%	Valid Error Empty 100%0%0%	Valid Error Empty 100%0%0%	Valid Error Empty 100%0%0%	Valid Error Empty 100%0%0%
1	21	Male	Other	Undergraduate	Facebook	Networking	
2	23	Female	Other	Graduate	LinkedIn	Education	
3	22	Male	Canada	Undergraduate	Instagram	Entertainment	
4	18	Male	Other	High School	Snapchat	Entertainment	
5	24	Female	Other	Graduate	Facebook	Networking	
6	19	Female	Other	Undergraduate	Twitter	News	
7	22	Female	Other	Undergraduate	Instagram	Entertainment	
8	21	Female	Other	Undergraduate	Facebook	Networking	
9	24	Male	USA	Graduate	YouTube	Education	
10	19	Male	Other	Undergraduate	TikTok	Entertainment	
11	18	Male	Other	High School	LinkedIn	Education	
12	18	Male	Other	High School	Snapchat	Entertainment	
13	20	Male	Other	Undergraduate	Facebook	Networking	
14	23	Female	Other	Graduate	LinkedIn	Education	
15	24	Male	India	Graduate	Instagram	Entertainment	
16	18	Female	Other	High School	Twitter	News	
17	22	Female	USA	Undergraduate	YouTube	Education	
18	21	Male	Australia	Undergraduate	YouTube	Education	
19	21	Male	Canada	Undergraduate	Facebook	Networking	
20	20	Male	Other	Undergraduate	Instagram	Entertainment	
21	23	Female	Other	Graduate	Facebook	Networking	
22	19	Female	Other	Undergraduate	Snapchat	Entertainment	
23	24	Male	Other	Graduate	Facebook	Networking	
24	23	Male	India	Graduate	TikTok	Entertainment	
25	22	Female	Other	Graduate	LinkedIn	Education	

DATA MODELING

Star Schema

FACT TABLE

Student_Data

DIMENSION TABLES

- Dim_Country
- Dim_Gender
- Dim_Platform
- Dim_Academic_Level
- Dim_Stress_Level

RELATIONSHIPS

One-to-Many Relationships

FILTER DIRECTION

Single Direction

MODEL BENEFITS

- Improved Performance
- Better Analysis
- Easy DAX Development
- Scalable Reporting

Dim_Work_Arrangement

RELATIONSHIPS

One-to-Many Relationships

FILTER DIRECTION

Single Direction

MODEL BENEFITS

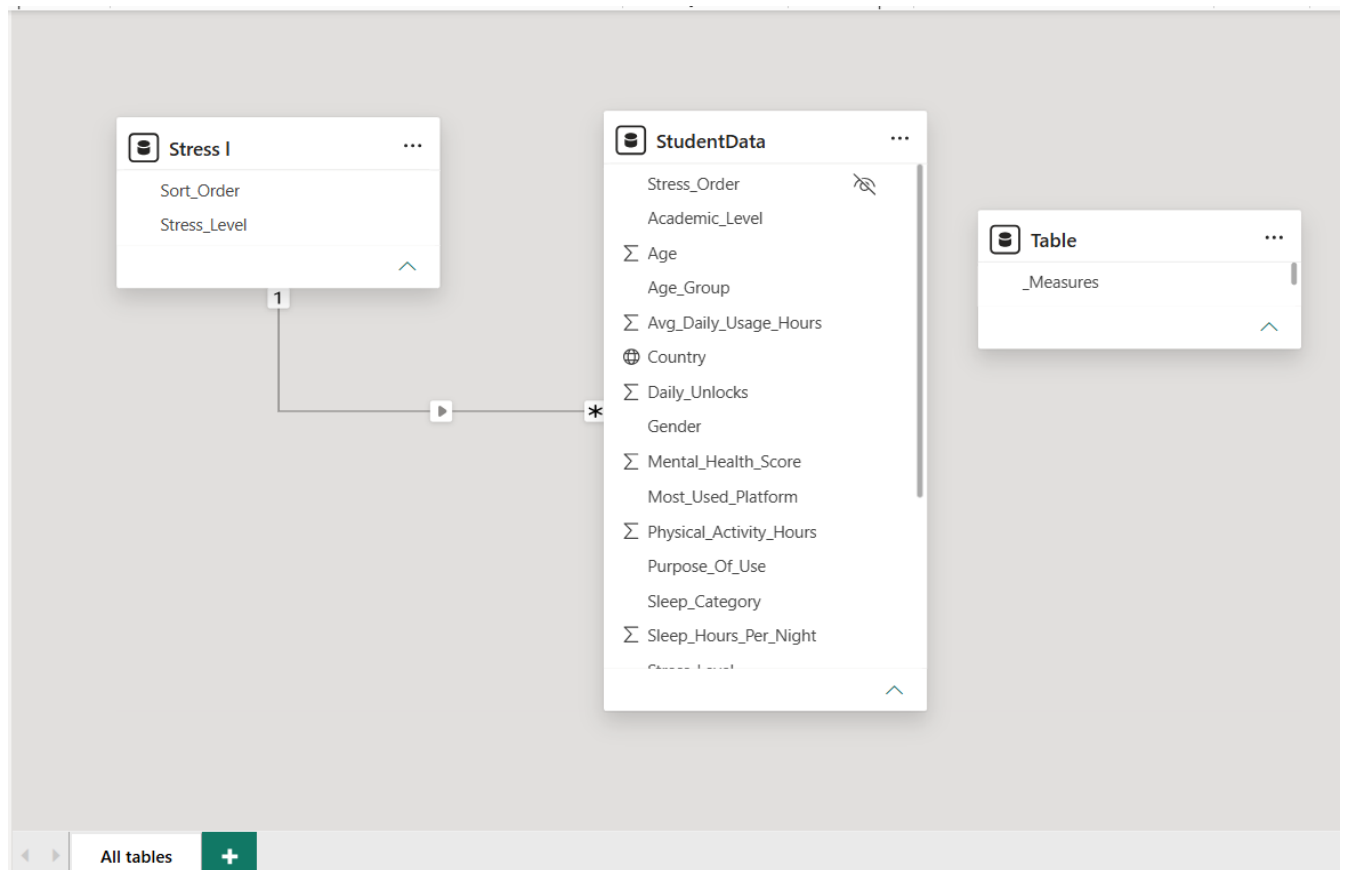
Improved Performance

Easy DAX Development

Better Scalability

SCREENSHOT:

Model View:



DAX MEASURES USED

Total Students

Total Students = COUNTROWS(Student_Data)

Purpose: Calculate total students.

Avg Mental Health Score

Avg Mental Health Score =
AVERAGE(Student_Data[Mental_Health_Score])

Purpose: Calculate average mental health score.

Avg Daily Usage Hours

Avg Daily Usage Hours =
AVERAGE(Student_Data[Daily_Usage_Hours])

Purpose: Calculate average social media usage.

Avg Stress Numeric

Avg Stress Numeric =
AVERAGE(Student_Data[Stress_Numeric])

Purpose: Measure average stress level.

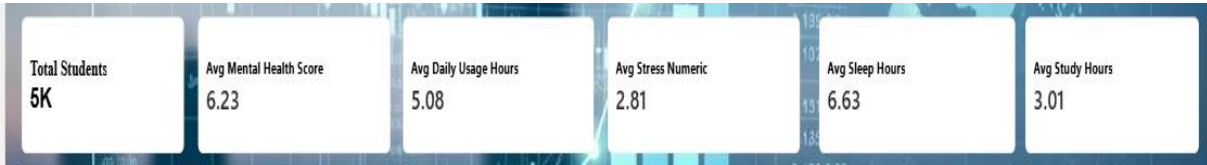
Avg Sleep Hours

Avg Sleep Hours =
AVERAGE(Student_Data[Sleep_Hours])

Purpose: Measure average sleep duration.

VISUALIZATIONS USED

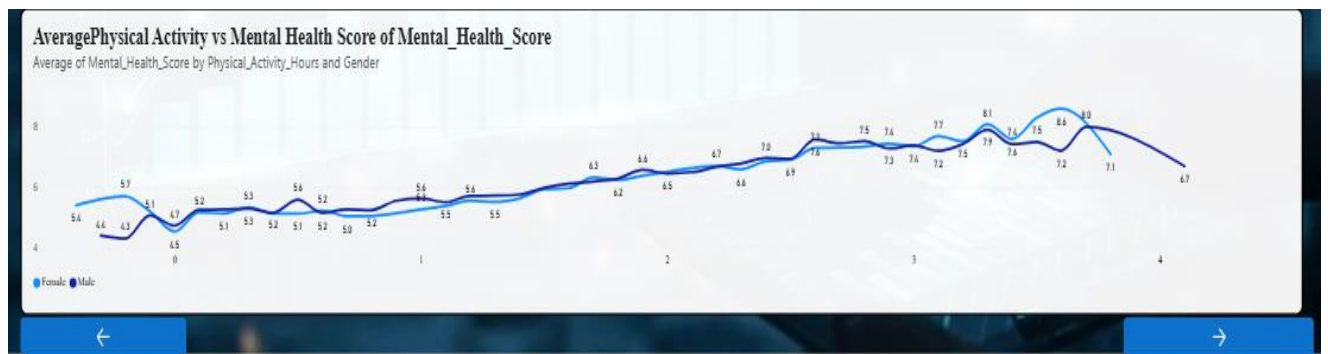
KPI CARDS:



BAR CHART:



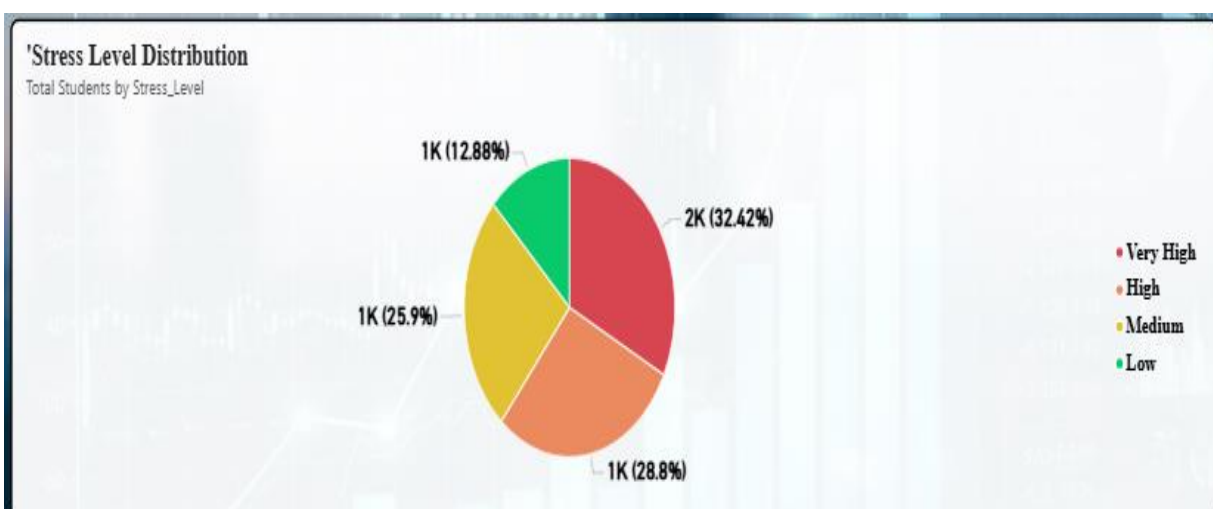
LINE CHART: .



MAP:



Donut CHART:



DASHBOARD SCREENSHOTS

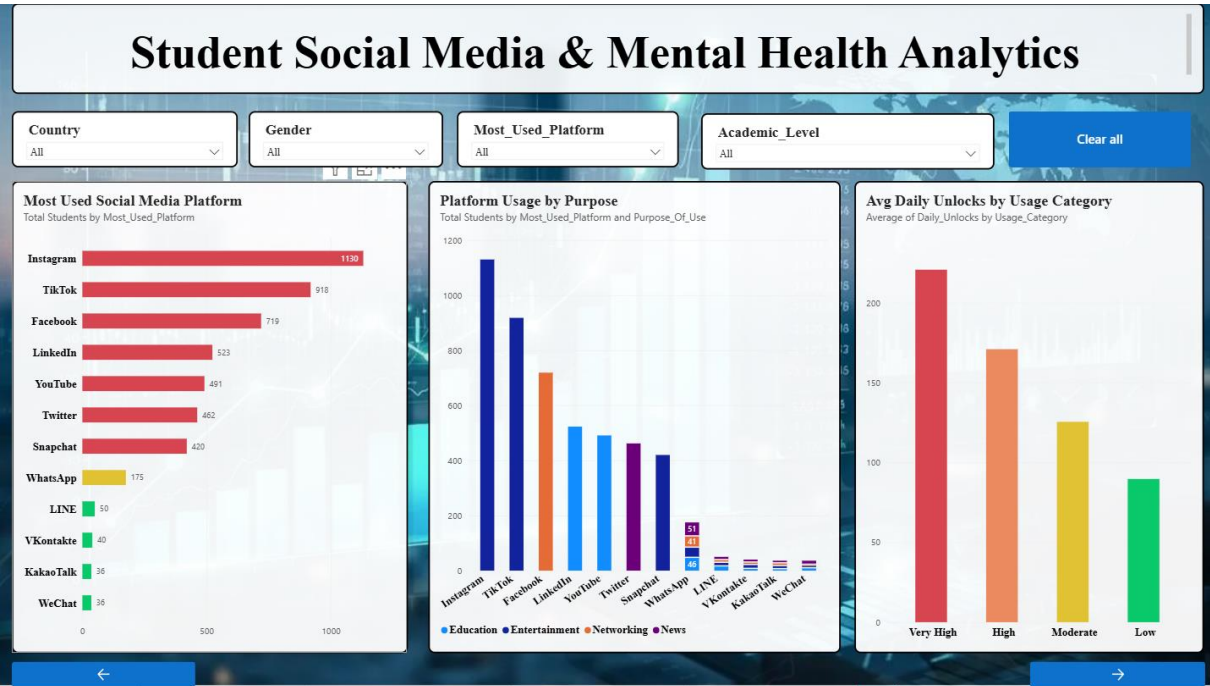
EXECUTIVE OVERVIEW



Explanation

Provides overall student mental health performance, stress distribution, platform usage and country-wise analysis.

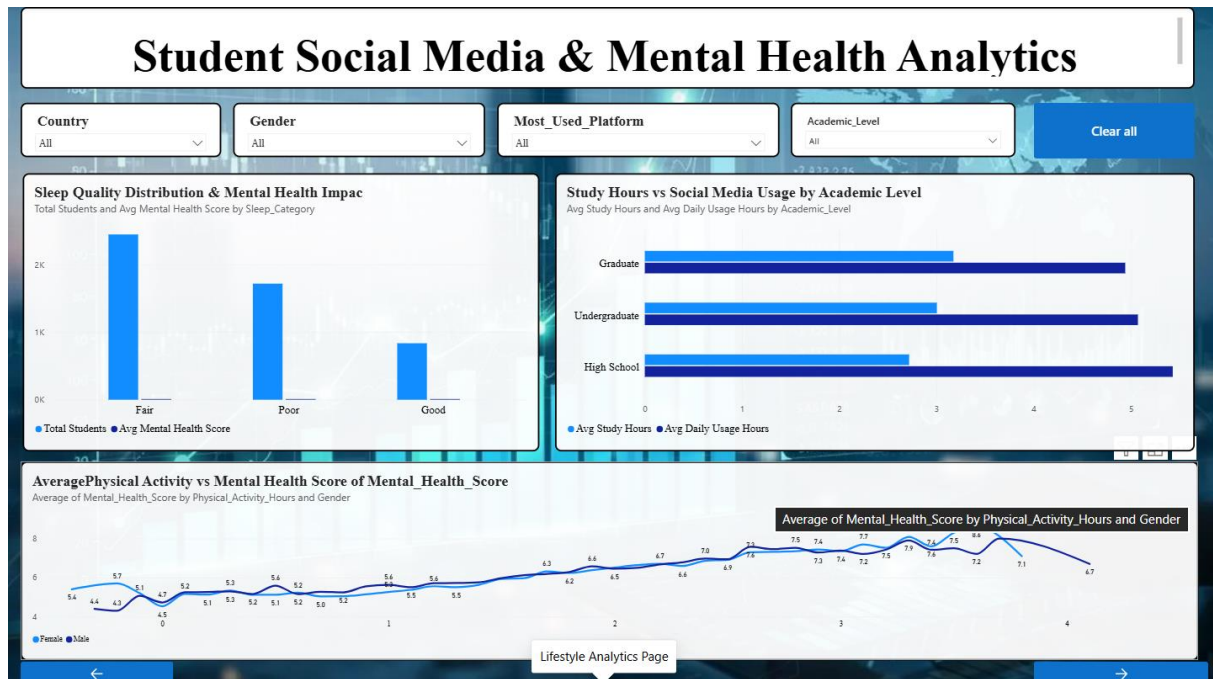
Social Media Analytics



Explanation

Analyzes platform popularity, purpose of usage and daily unlock behavior.

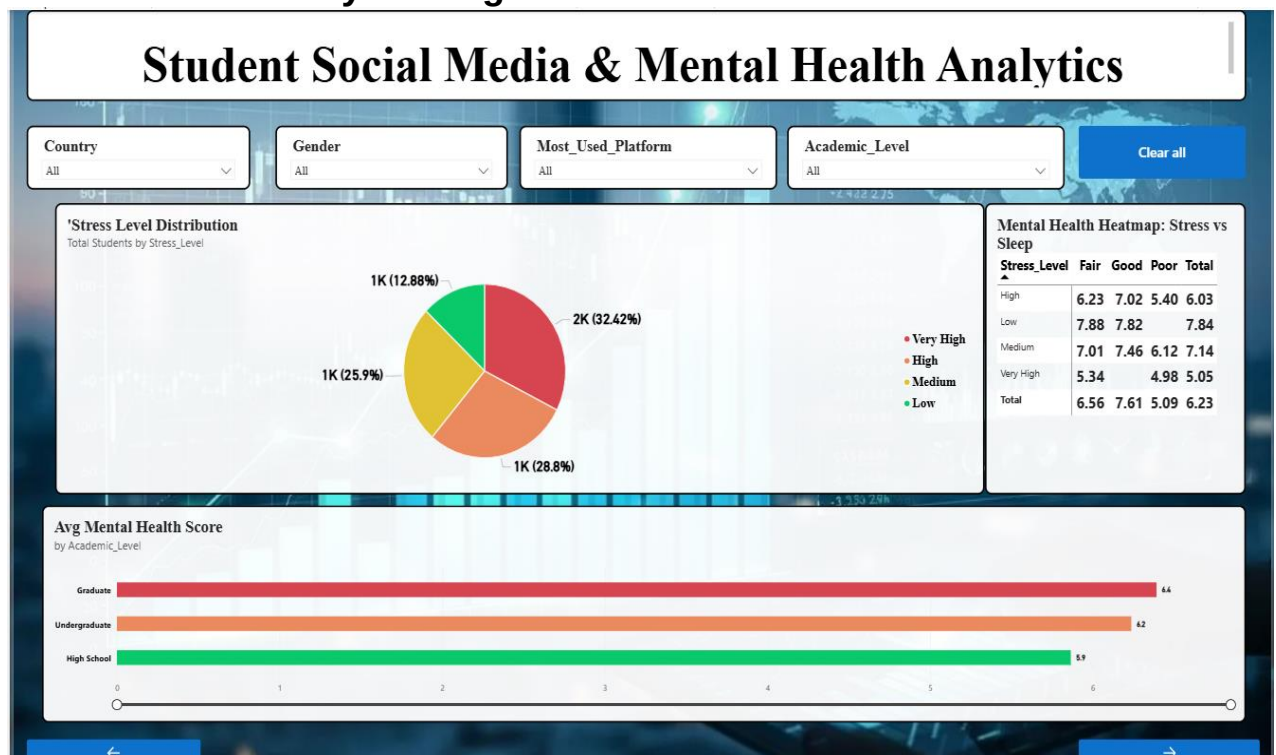
Lifestyle Analytics



Explanation:

Shows relationship between sleep quality, study hours, physical activity and mental health.

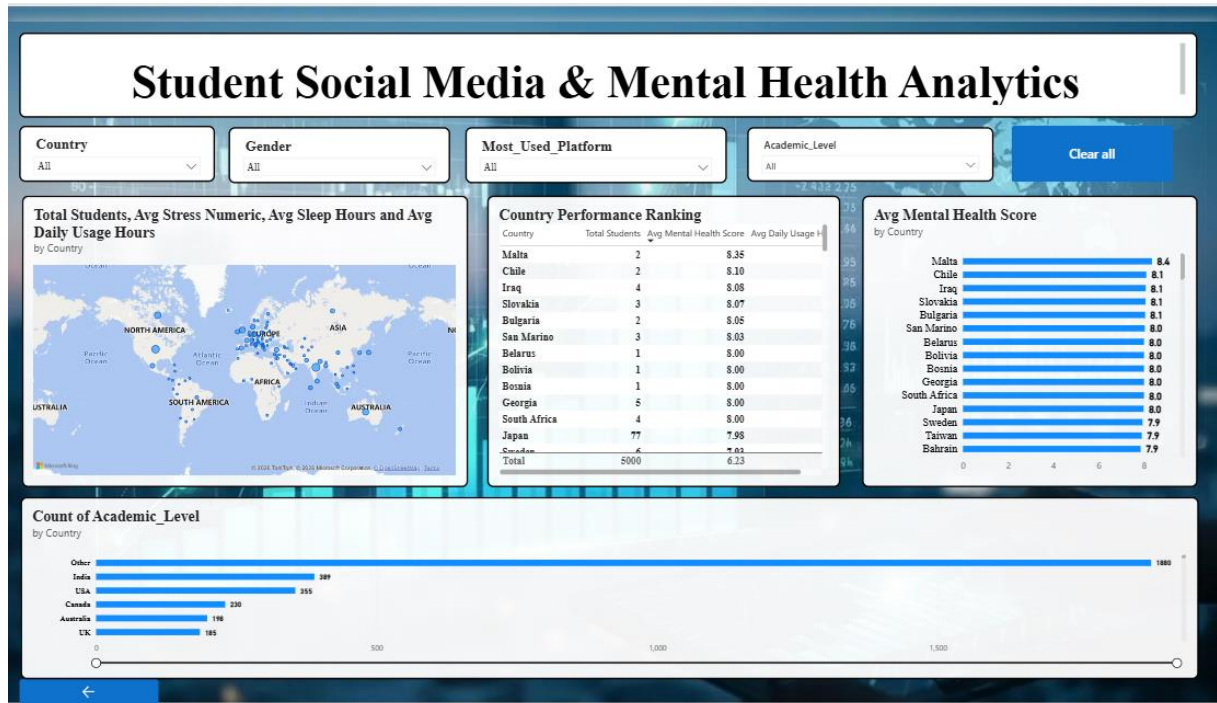
Mental Health Analytics Page



Explanation:

Displays stress level distribution and sleep quality impact.

Country Analytics Dashboard



Explanation:

Ranks countries based on mental health scores and student participation.

KEY INSIGHTS

INSIGHT-1

Instagram is the most used social media platform among students

INSIGHT-2

Students with better sleep quality tend to have higher mental health scores.

INSIGHT-3

Higher physical activity is associated with improved mental health.

INSIGHT 4

Very High stress category contains the largest student group.

BUSINESS RECOMMENDATIONS

RECOMMENDATION-1

Promote healthy social media usage habits.

RECOMMENDATION-2

Encourage better sleep schedules.

RECOMMENDATION-3

Increase awareness of mental health support programs.

RECOMMENDATION-4

Promote physical activity among students.

FEATURES IMPLEMENTED

- ✓ Drill Down
- ✓ Drill Through
- ✓ Tooltips
- ✓ Dynamic Titles
- ✓ Slicers
- ✓ Conditional Formatting
- ✓ Navigation Buttons

CHALLENGES FACED

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- Data Cleaning Challenges
- Relationship Management
- DAX Optimization
- Dashboard Layout Design
- Performance Tuning

LEARNINGS

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- Power Query
- Data Modeling
- DAX Development
- Dashboard Design
- Storytelling
- Business Analysis

PROJECT STORYTELLING

BUSINESS SCENARIO

Educational institutions want to understand how social media usage impacts student mental health and academic performance.

KEY FINDINGS

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- Instagram dominates platform usage.
- Better sleep improves mental health.
- Higher physical activity improves mental health outcomes.
- High social media usage correlates with increased stress

BUSINESS IMPACT

- Better student support systems.
- Improved awareness programs.
- Data-driven educational decisions.
- Enhanced student well-being initiatives.

PROJECT DEPLOYMENT

DEPLOYMENT OBJECTIVE

Deploy the Student Social Media & Mental Health Analytics Dashboard to enable educators, researchers, and stakeholders to access interactive reports and insights related to student social media usage, mental health, stress levels, sleep quality, and academic behavior in real time.

DEPLOYMENT PLATFORM

Power BI Service

DEPLOYMENT PROCESS

Power BI Service

STEP-1

Validate dataset quality, data model relationships, and DAX measures in Power BI Desktop.

STEP-2

Optimize dashboard design, visual performance, navigation buttons, and user experience.

STEP-3

Publish the PBIX file to Power BI Service.

STEP-4

Configure dataset refresh settings to ensure updated analytics.

STEP-5

Create dashboards and reports from the published dataset.

STEP-6

Apply workspace permissions and secure stakeholder access.

Share dashboards with educators, researchers, and project reviewers.

BUSINESS BENEFITS

- Centralized Student Analytics Platform
- Interactive Mental Health Monitoring
- Social Media Usage Insights
- Real-Time Student Behavior Analysis
- Improved Educational Decision-Making
- Better Student Well-Being Assessment
- Data-Driven Research and Reporting
- Enhanced Awareness of Mental Health Trends

CONCLUSION

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The Student Social Media & Mental Health Analytics Dashboard successfully analyzed student behavior, social media usage patterns, stress levels, sleep quality, study habits, physical activity, and overall mental health outcomes. By integrating multiple dimensions of student lifestyle and academic factors, the dashboard provided a comprehensive view of the relationship between social media engagement and student well-being. The project generated meaningful and actionable insights that can help educational institutions, researchers, and policymakers identify key factors affecting mental health and academic performance. These insights can support the development of awareness programs, wellness initiatives, and data-driven strategies aimed at improving student well-being, promoting healthy digital habits, enhancing academic success, and fostering a healthier learning environment for students across different educational levels and regions.